

Asvanth C

Software Engineer

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Software engineer with two years of production experience at a digital signage company, working across three layers of the same product: backend services in Java and the Play Framework, the OAuth 2.0 and OpenID Connect identity layer that guards them, and the player software running on Samsung, LG and Android displays in the field.

EXPERIENCE

Software Engineer · Zeetaminds Aug 2024 – Present · Coimbatore, India

Digital signage platform: a content management system that schedules and delivers media to displays running in retail, offices, campuses and public spaces.

- Backend.** Build and extend REST APIs in Java on the Play Framework, using Guice for dependency injection and MyBatis over a relational schema, serving both the management web client and the device agents in the field.
- Identity and access management.** Work on OAuth 2.0 and OpenID Connect flows — Authorization Code with PKCE, token exchange and single sign-on — and on JWT verification: RS256 signatures, JWKS key distribution, issuer, audience and expiry validation, refresh and session lifetime, login CSRF protection via state, and nonce.
- Device platforms.** Build and debug the JavaScript player application running on Samsung Tizen (SSSP), LG webOS and Android signage devices, behind a thin platform abstraction with a separate implementation per vendor.
- Media playback.** Diagnose playback failures across container, codec, profile and level, and hardware decoder limits, including silent failures where a device reports no error and renders black.
- Device management.** Work on remote control, snapshots and live view, using acknowledged commands and explicit timeouts against devices whose state drifts continuously.
- Web client and compatibility.** Contribute features to the Angular and TypeScript management client, and design API changes to be additive and backward compatible: devices in the field run mixed firmware and cannot be assumed to update.

PROJECTS

Portfolio Dashboard Node.js · Express · React · Vite

- Replaced a spreadsheet with a dashboard served by an Express JSON API. Rewrote price fetching to be bulk-first across a tiered provider fallback chain after the original per-ticker approach was rate-limited into failing, with cache lifetimes matched to how quickly each source actually changes.

Personal site and engineering notes Astro · MDX · GitHub Actions

Ten long-form engineering notes on OAuth 2.0 and OpenID Connect, JWT verification, media playback across device platforms and production debugging — asvanthc.github.io/writing

SKILLS

Languages:	Java, JavaScript, TypeScript, SQL, Python, C/C++
Backend:	Play Framework, REST API design, MyBatis, Guice, Node.js, Express
Identity:	OAuth 2.0, OpenID Connect, Authorization Code with PKCE, JWT, RS256, JWKS, SSO, session and token lifecycle
Web:	Angular, TypeScript, HTML, CSS, browser APIs, embedded web runtimes
Platforms:	Samsung Tizen (SSSP), LG webOS, Android, Linux
Data & tools:	MySQL, schema and query design, MongoDB, Git, Gradle, Maven, Nginx
Learning:	Docker, CI/CD, LLM APIs and tool calling, RAG, distributed systems

EDUCATION

B.Tech, Computer Science and Engineering · Amrita Vishwa Vidyapeetham 2020 – 2024 · GPA 8.18/10

Coimbatore, Tamil Nadu. Coursework in data structures and algorithms, operating systems, databases, computer networks, distributed computing, computer architecture, and cryptography and network security.

CERTIFICATIONS

21 edX certificates (2019–2022) from AWS, IBM, The Linux Foundation, Columbia University and the University of Washington — cloud development, containers and Kubernetes, data science and cybersecurity. Credential IDs at asvanthc.github.io.